

Employee Management MVC

ASP.NET Core MVC • C# • Entity Framework Core • SQL Server • Bootstrap

1. Project Overview

Employee Management MVC is a web application used to manage employees and departments. It is built using ASP.NET Core MVC. The application demonstrates the basic structure of an MVC project, CRUD operations, form validation, database connectivity, Razor Views, Bootstrap and Entity Framework Core.

2. Main Features

- Dashboard for a simple overview of the application.
- Employee module to add, view, edit and delete employee records.
- Department module to add, view, edit and delete departments.
- Data Annotation validation for employee and department forms.
- Responsive design using Bootstrap.
- Shared navigation and common page layout.
- Razor Views for displaying application pages.
- Entity Framework Core for communication with SQL Server.
- EF Core migrations for creating and updating the database.
- Custom error handling for invalid or unavailable pages.

3. Technologies Used

Technology	Purpose
C#	Application programming language
ASP.NET Core MVC	Builds the web application
Entity Framework Core	Connects the application to the database
SQL Server	Stores employee and department data
Razor Views	Creates dynamic web pages
Bootstrap	Provides responsive and clean UI
Git & GitHub	Version control and project sharing

4. MVC in Simple Words

MVC means Model, View and Controller. These three parts keep the application organized.

- Model – Represents application data. Example: Employee and Department classes.

- View – The web page shown to the user. Razor .cshtml files are used for Views.
- Controller – Receives the user's request, performs the required operation and returns a View.

Example: When the user clicks Employees, the request goes to EmployeeController. The controller gets employee data from the database and sends it to the Employee View.

5. Project Folder Structure

- Controllers/ – Contains controllers such as HomeController, EmployeeController and DepartmentController.
- Models/ – Contains Employee.cs and Department.cs.
- Data/ – Contains ApplicationDbContext, which connects the application to SQL Server.
- Views/ – Contains Razor pages for Home, Employee, Department and Shared layout.
- Views/Shared/ – Contains common files such as _Layout.cshtml.
- Migrations/ – Contains Entity Framework database migration files.
- wwwroot/ – Contains CSS, JavaScript and other static files.
- Properties/ – Contains project launch settings.
- Program.cs – Configures services, Entity Framework and MVC routing.
- appsettings.json – Contains application settings and the database connection string.
- .gitignore – Prevents temporary and local files from being uploaded to GitHub.

6. Employee Module

The Employee module manages employee information. Each employee can have a first name, last name, email, phone number, salary, joining date and department.

- List – Displays employees.
- Details – Displays complete information for one employee.
- Create – Adds a new employee.
- Edit – Updates an existing employee.
- Delete – Removes an employee after confirmation.

7. Department Module

The Department module manages departments such as HR, Finance, IT or Sales. An employee can be assigned to a department.

- List – Displays departments and the number of employees.
- Create – Adds a new department.
- Edit – Updates department information.
- Delete – Removes a department when it is not being used by employees.

8. Data Validation

Data Annotations are used to check user input before saving it. For example, an employee email must have a valid email format and salary must be within the allowed range.

- [Required] – Makes a field mandatory.

- [StringLength] – Limits the number of characters.
- [EmailAddress] – Checks email format.
- [Phone] – Checks phone number format.
- [Range] – Limits numeric values.
- [DataType(DataType.Date)] – Treats a value as a date.

9. Entity Framework Core and Database

Entity Framework Core (EF Core) is used to work with SQL Server without writing every SQL query manually. ApplicationDbContext contains DbSet properties for Employees and Departments.

Database setup commands:

- dotnet ef migrations add InitialCreate
- dotnet ef database update
- Migrations keep the C# model and database structure synchronized. If the model changes, a new migration can be created and applied to the database.

10. Application Flow

- User opens a page in the browser.
- MVC routing identifies the correct controller and action.
- The controller receives the request.
- The controller uses the model and ApplicationDbContext when database data is required.
- Entity Framework Core communicates with SQL Server.
- The controller sends the result to a Razor View.
- The View displays the result to the user.